INTAKE SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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Note
- If the intake shutter valve is replaced, perform the “Operation After Replacing Intake Shutter Valve”.

Operation After Replacing Intake Shutter Valve

1. If the intake shutter valve is replaced, perform the following procedure.

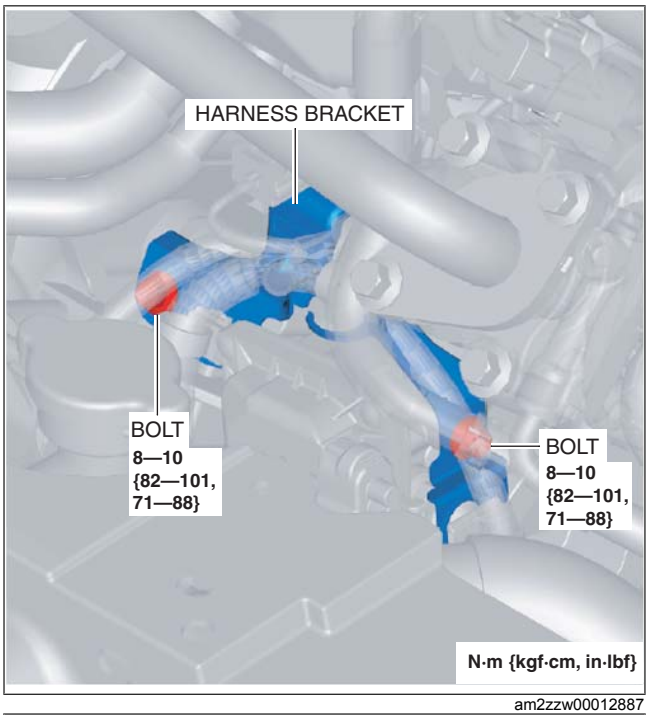
STEP	ACTION	PAGE/CONDITION
1	Perform intake shutter valve initialization procedure.	(See INTAKE SHUTTER VALVE INITIALIZATION [SKYACTIV-D 1.5].)
2	Switch the ignition off.	—
3	Wait for 20 s or more.	—
4	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
5	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)

Intake Shutter Valve Removal/Installation

Warning
- **Never remove the cooling system cap or loosen the radiator drain plug while the engine is running, or when the engine and radiator are hot. Scalding engine coolant and steam may shoot out and cause serious injury. It may also damage the engine and cooling system.**
- **Turn off the engine and wait until it is cool. Even then, be very careful when removing the cap. Wrap a thick cloth around it and slowly turn it counterclockwise to the first stop. Step back while the pressure escapes.**
- **When you are sure all the pressure is gone, press down on the cap using the cloth, turn it, and remove it.**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the following parts: (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - Fresh-air duct
   - Air cleaner
   - Air hose
   - Air intake pipe
3. Remove the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Set the DC-DC converter (i-ELOOP) aside so that it does not interfere with the servicing. (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5. Remove the turbocharger air outlet pipe. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

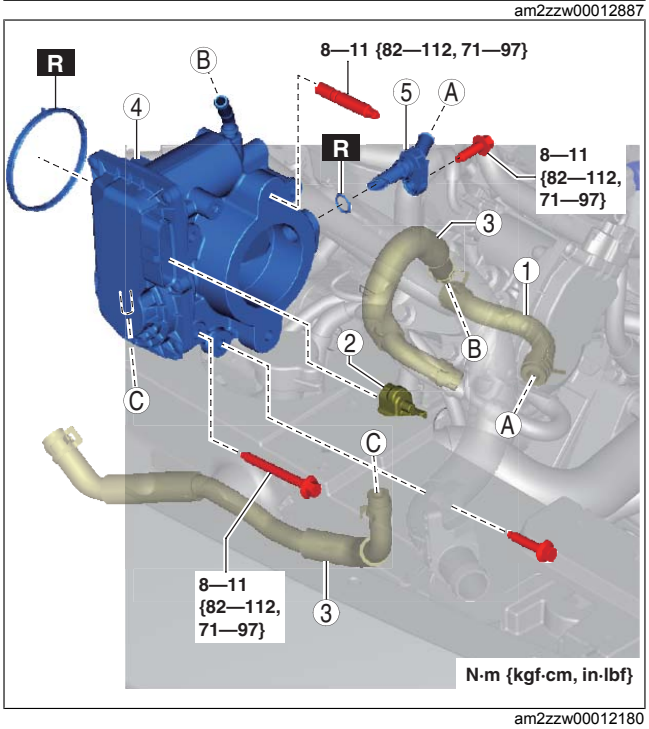
6. Remove the wiring harness bracket bolts shown in the figure.
7. Remove the cooling system cap.



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8. Remove in the order shown in the figure.

1	Vacuum hose (See Vacuum hose installation note.)
2	Intake shutter valve connector
3	Water hose (See Water hose removal note.) (See Water hose installation note.)
4	Intake shutter valve (See Operation After Replacing Intake Shutter Valve.)
5	Air pipe

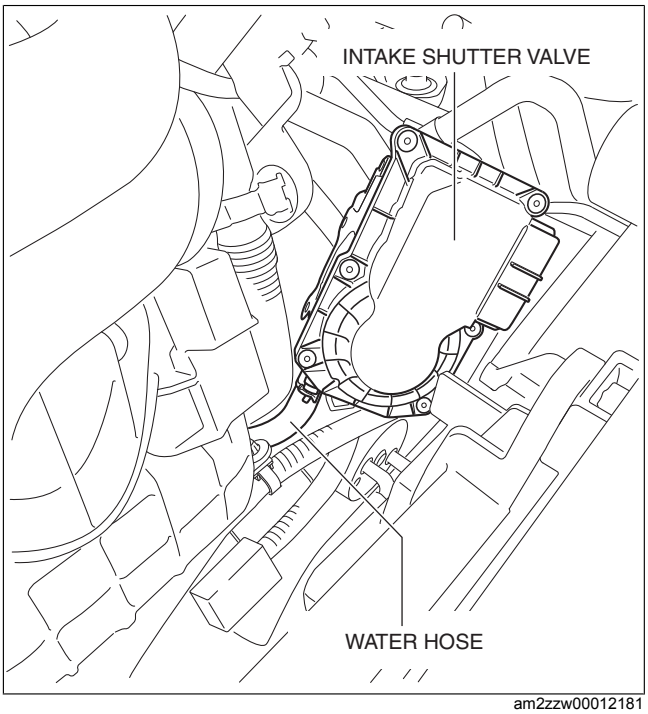


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9. Install in the reverse order of removal.
10. Refill the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)

Water hose removal note

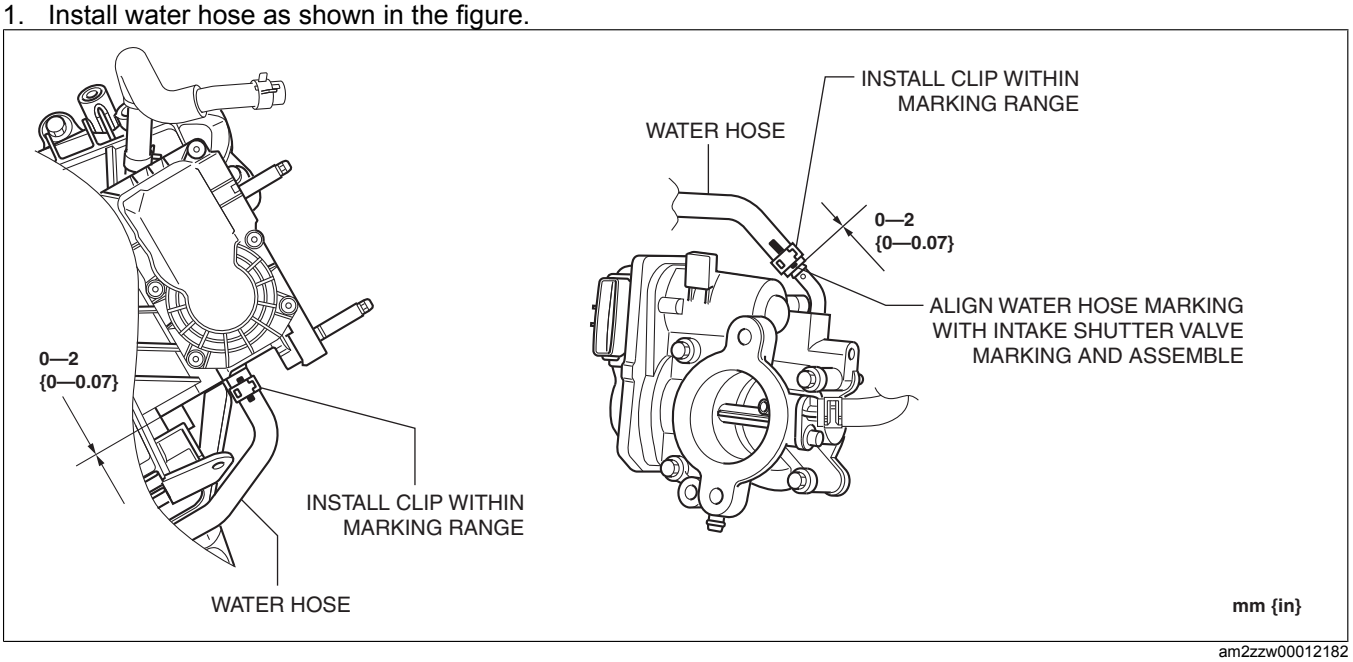
1. Move the intake shutter valve to the water hose disconnection position.
2. Disconnect the water hose from the intake shutter valve and plug the water hose quickly.



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Water hose installation note

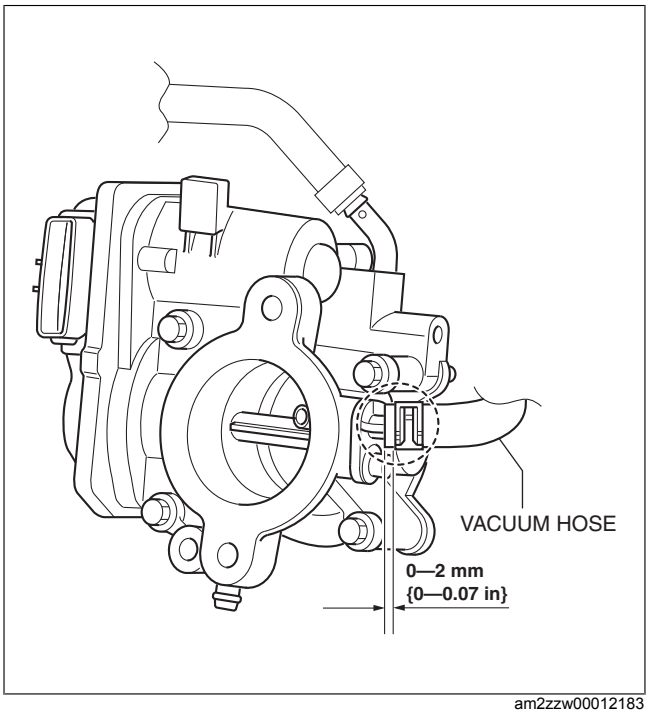
1. Install water hose as shown in the figure.



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Vacuum hose installation note

1. Install vacuum hose as shown in the figure.



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